

HIEGNet: A Heterogeneous Graph Neural Network Including the Immune Environment in Glomeruli Classification

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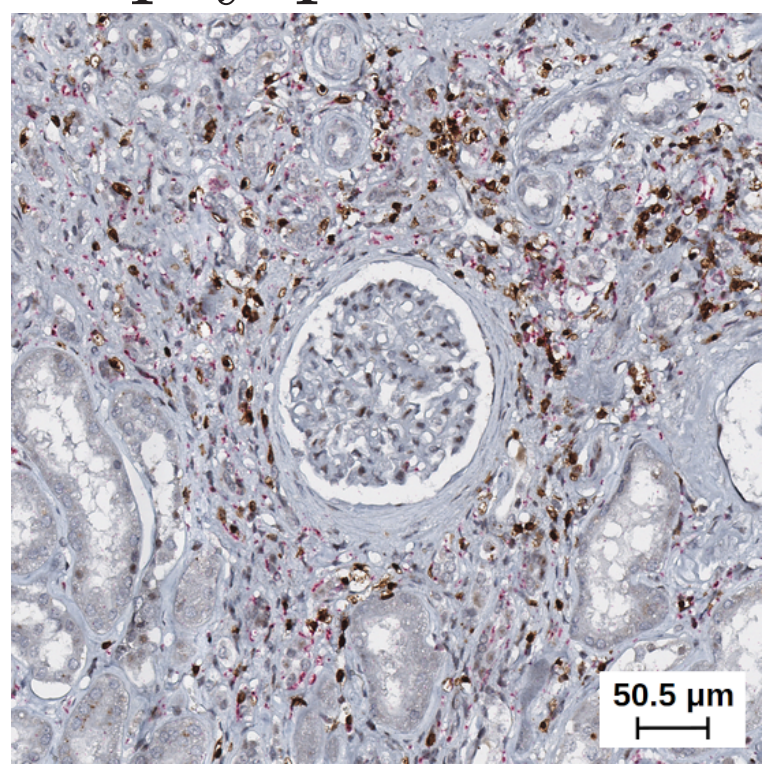
Summary

We present a graph-based approach for classifying glomerular health in histopathological images based on the immune environment.

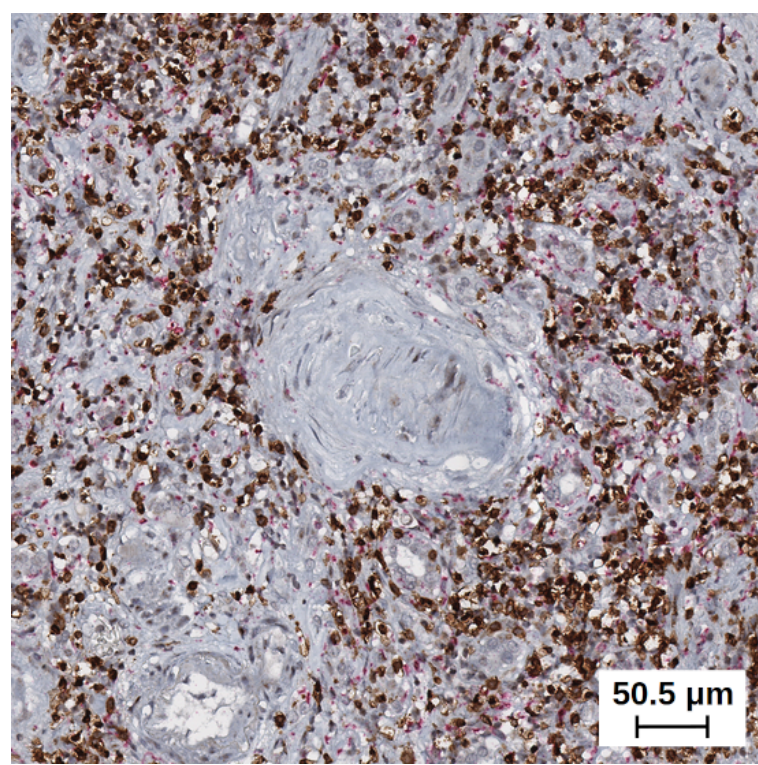
- We use a pipeline to create an attributed heterogeneous graph of immune cells and glomeruli from WSIs.
- We designed a Graph Neural Network that accounts for the edge types between the different node types.

Problem: Tissue Classification

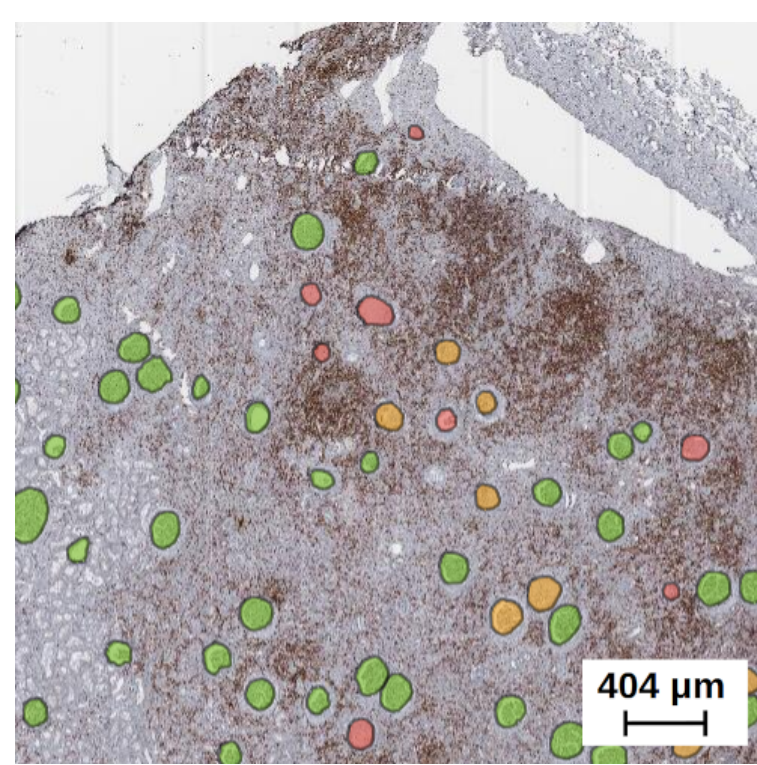
- *Glomeruli* (glom) are cell structures in the kidneys which filter blood.
- Goal: *classification of glomeruli* into “healthy”, “sclerotic” and “dead”.
- Aiding in early detection of kidney-transplant rejection symptoms in biopsy photos.



(a) Healthy glom



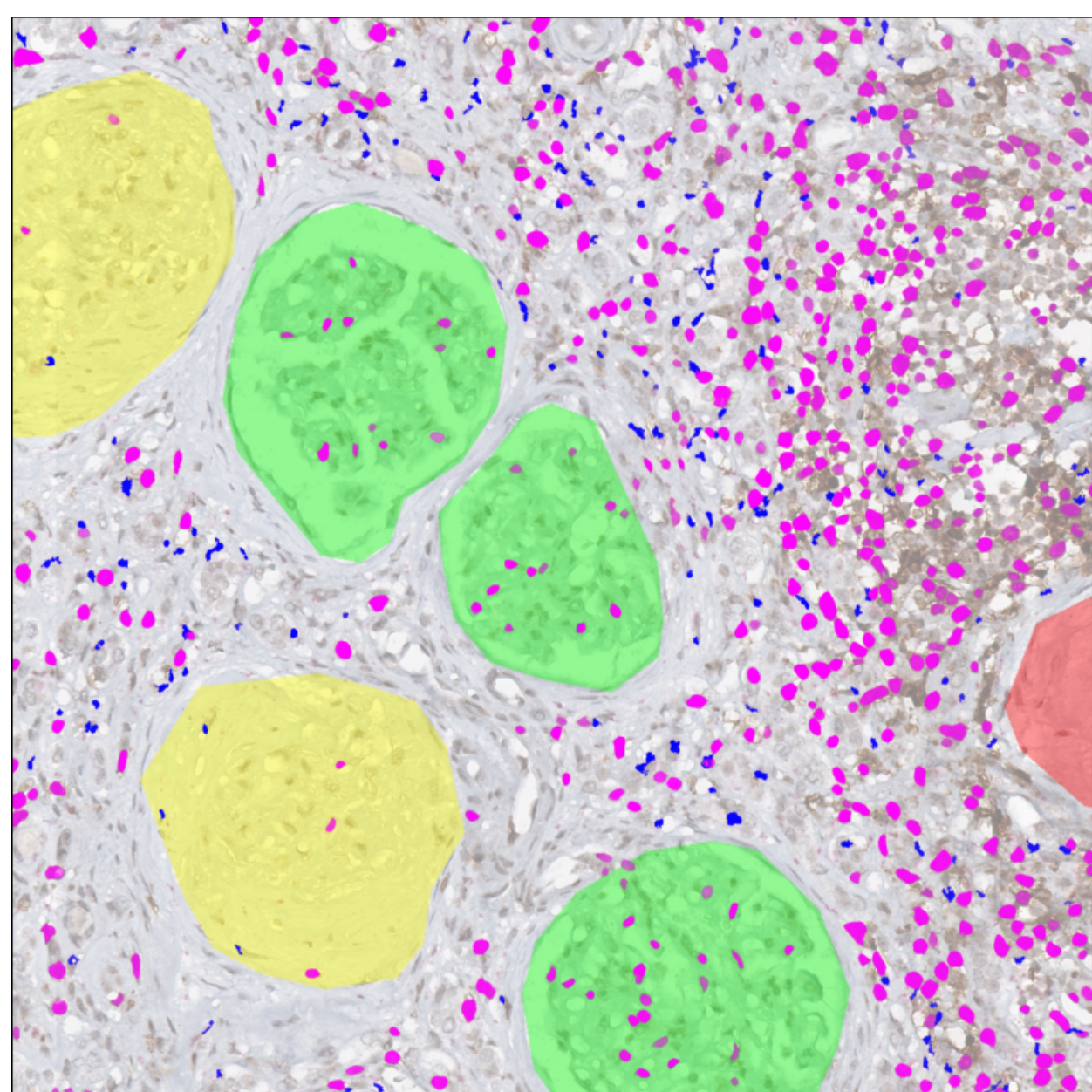
(b) Dead glom



(c) Tissue slice

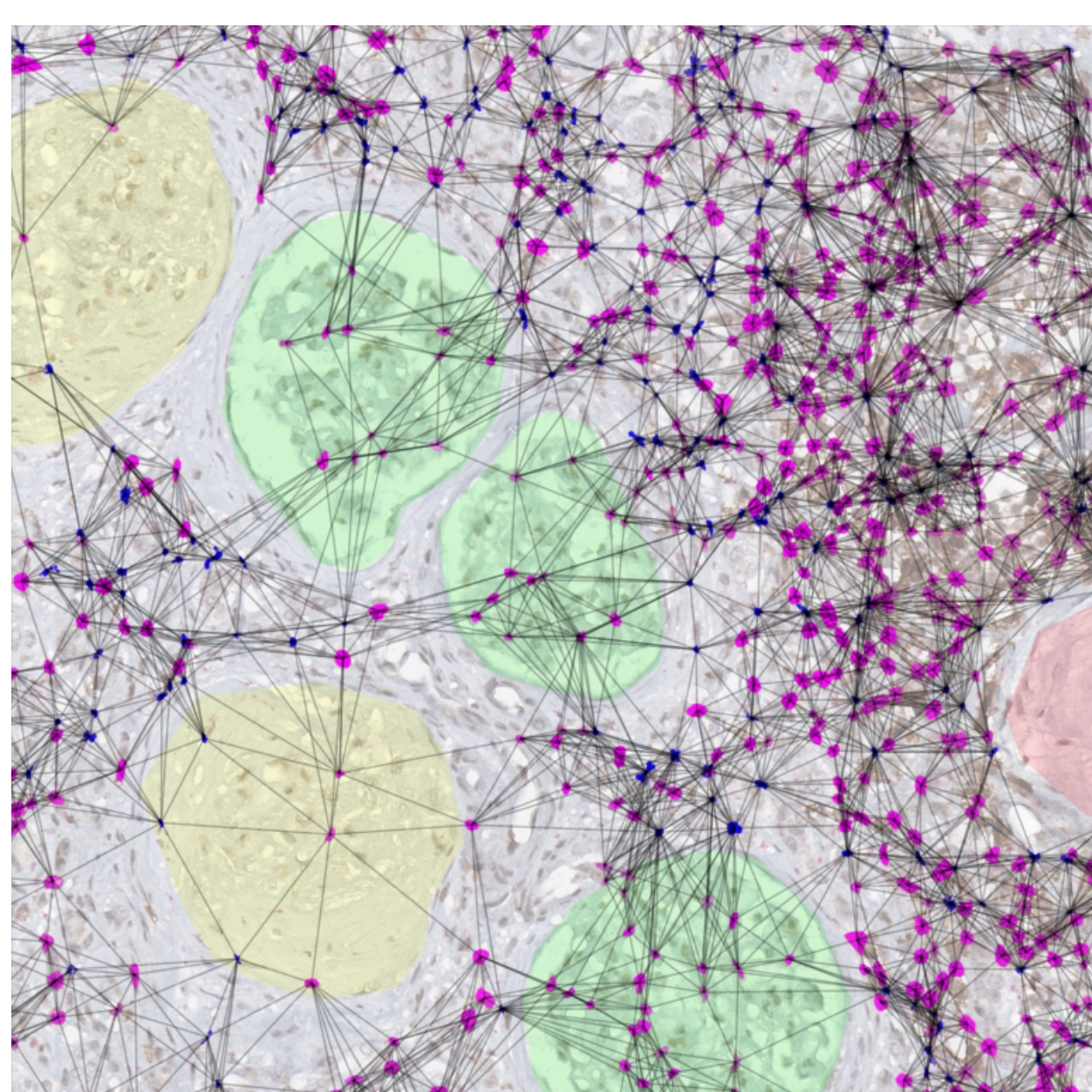
Graph Construction

Graph Nodes & Features

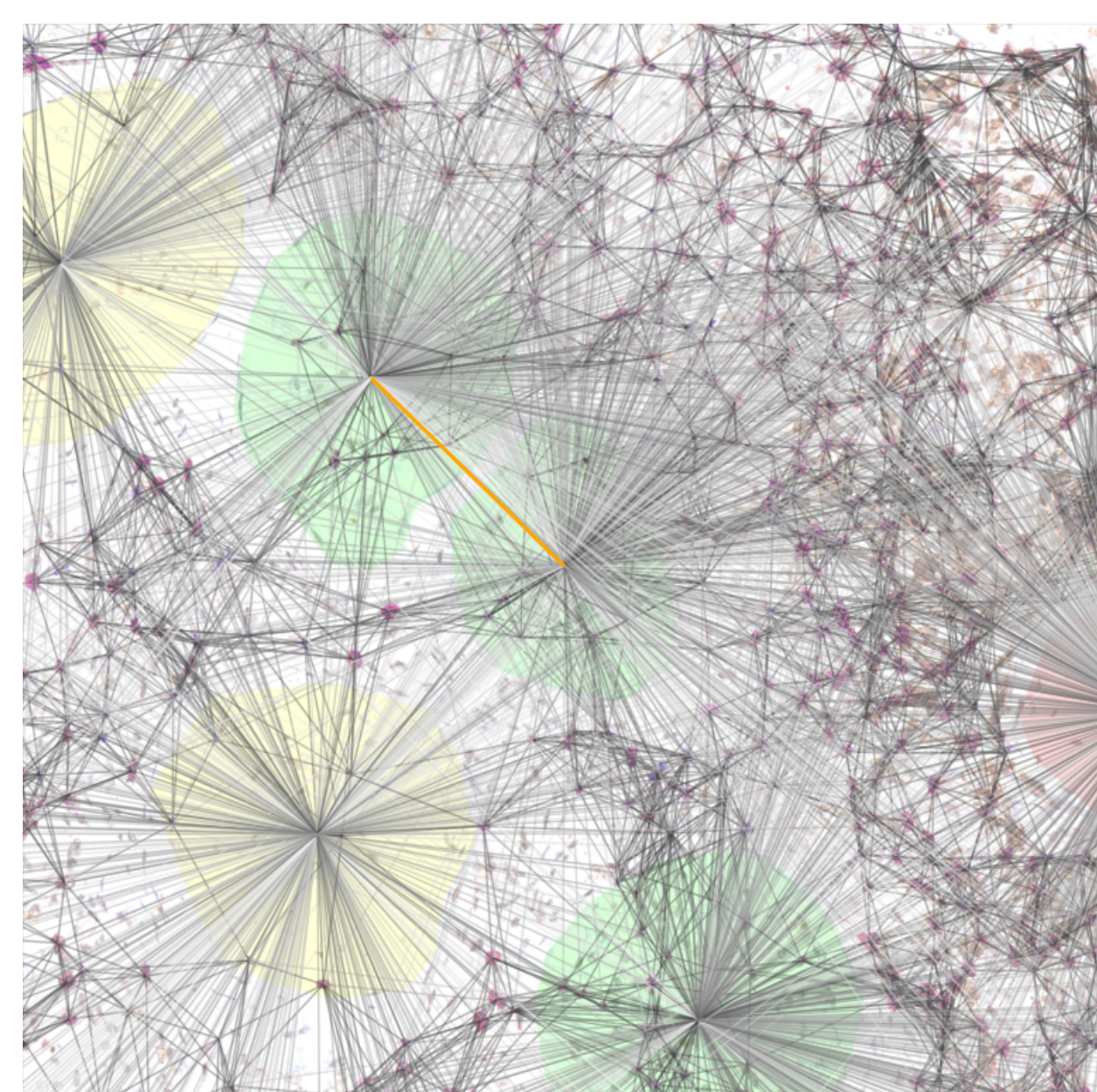
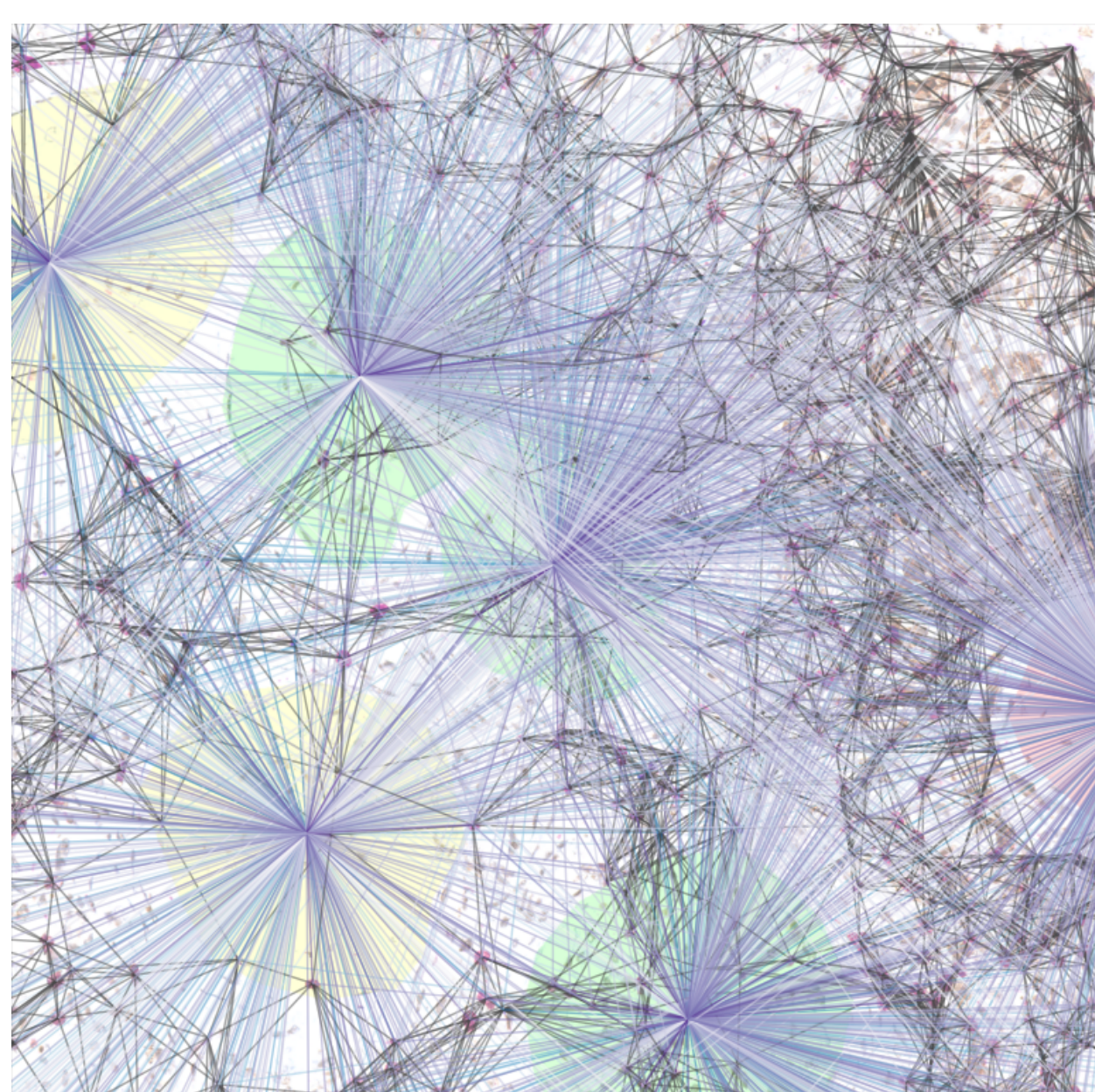


- Our graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T}, \mathcal{R})$ is heterogeneous, i.e., has different node types $\mathcal{T}$ and edge types $\mathcal{R}$.
- We included three node types $\mathcal{T} = \{g, m, t\}$ tissue structures:
 - Glomeruli g ,
 - Macrophages m ,
 - and T-cells t .
- We create initial node features $\mathbf{h}_v^{(0)}$ using Local Binary Patterns and shape-based features.

Graph Edges



- We constructed an edge $(v, r, u) \in \mathcal{E}$ depending on the Euclidean distance $d_{v,u}$.
- Three groups of edge types:
 - Between immune cells $\mathcal{R}_i = \{r_{m,t}, r_{t,m}, r_{m,m}, r_{t,t}\}$,
 - Between immune cells and close glomeruli $\mathcal{R}_{ig} = \{r_{g,t}, r_{g,m}, r_{m,g}, r_{t,g}\}$,
 - Between glomeruli $r_{g,g}$.



Proposed Architecture

Graph Neural Networks

The node features are updated by aggregating messages from neighbouring nodes via edges:

$$\mathbf{h}_v^{(\ell+1)} = U(\mathbf{h}_v^{(\ell)}, M(\mathbf{h}_v^{(\ell)}, \{\mathbf{h}_u^{(\ell)} : u \in \mathcal{N}(v)\})),$$

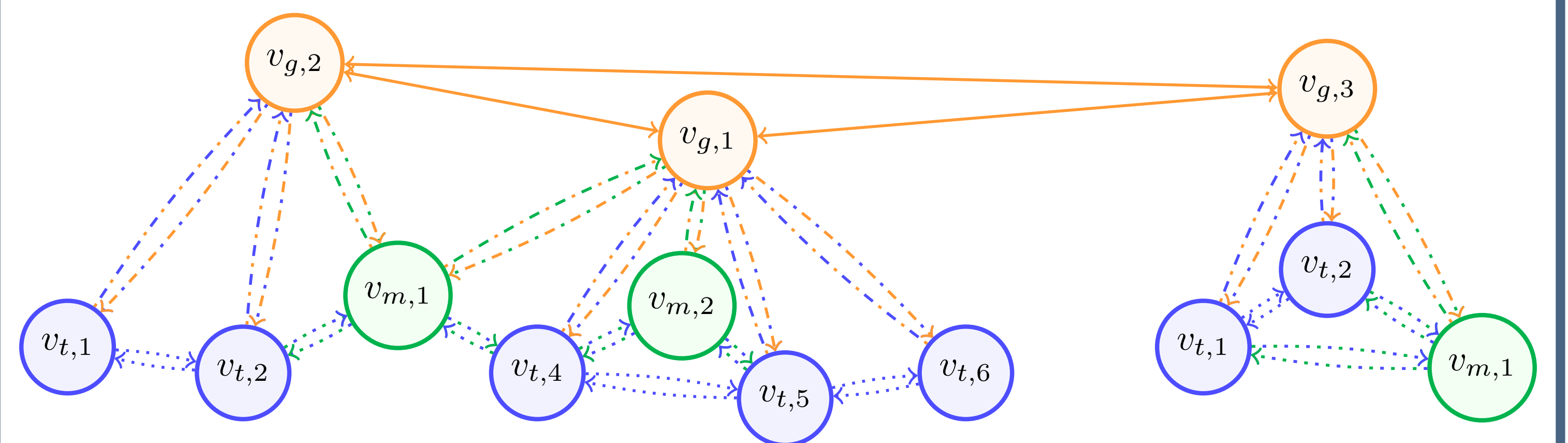
where $M(\cdot)$ and $U(\cdot)$ are arbitrary learnable message and update functions, $\mathbf{h}_v^{(\ell)}$ is the hidden representation of v at layer ℓ and $\mathcal{N}(v)$ is the set of neighbours of node v . The final node representations $\mathbf{h}_v^{(L)}$ are obtained after L layers, which are used for node classification [1].

HIEGNet

We define a HIEGNet message passing layer based on the Relational Graph Convolutional Network (RGCN) [2]:

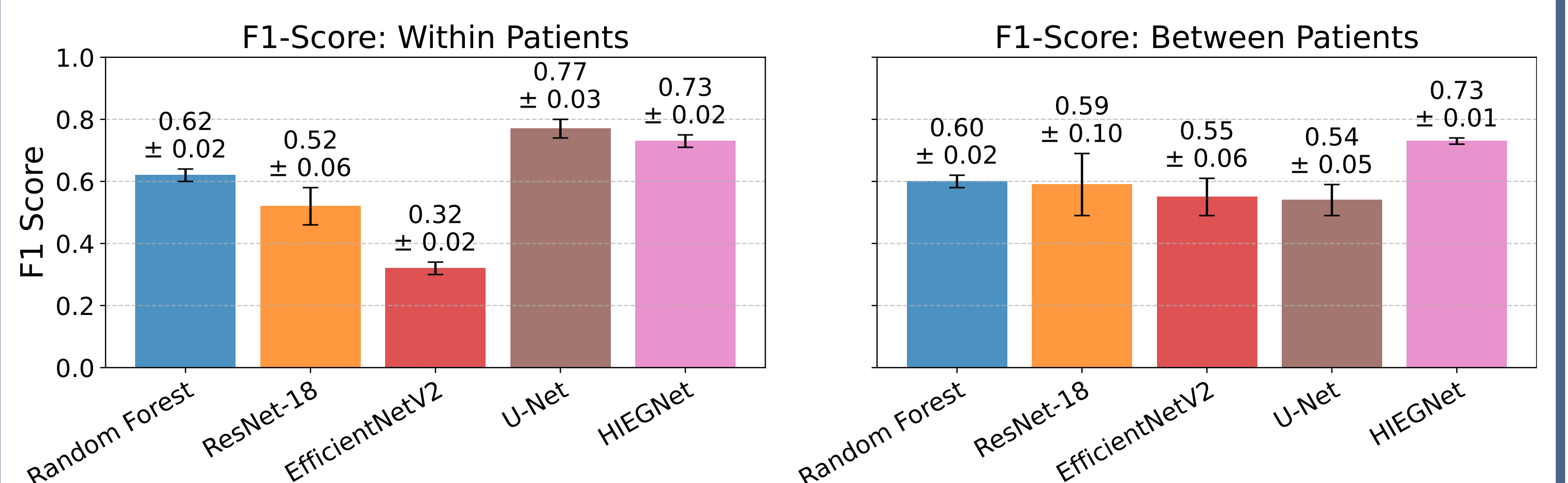
$$\mathbf{h}_v^{(\ell+1)} = \sum_{r \in \mathcal{R}} U_r^{(\ell)}(\mathbf{h}_v^{(\ell)}, M_r^{(\ell)}(\mathbf{h}_v^{(\ell)}, \{\mathbf{h}_u^{(\ell)} : (v, r, u) \in \mathcal{E}\})),$$

where $U_r^{(\ell)}(\cdot)$ and $M_r^{(\ell)}(\cdot)$ are update and message functions with a separate parameter set for each edge type r and layer ℓ . We define three specific message passing methods to align with the characteristics of the different edge type groups: $\{r_{g,g}\}$, $\mathcal{R}_i$, and $\mathcal{R}_{ig}$. The optimal message passing methods are identified through a grid search.



Experiments

We evaluated the performance of our model in two settings: (1) training and testing on a set of nodes from the same patients (within patients), (2) training and testing on different patients (between patients).



Further Directions

Generalisation: We see further potential in the development of our approach to become more robust to changes in the image domain due to varying lab conditions.

Explainability: To aid in the adoption of our approach, we also recommend future work on explainability methods.

References

- [1] J. Gilmer, S. S. Schoenholz, P. F. Riley, O. Vinyals, G. E. Dahl. “Neural Message Passing for Quantum Chemistry”, *ICLR*, 2017.
- [2] M. Schlichtkrull, T. N. Kipf, P. Bloem, R. van den Berg, I. Titov, M. Welling, “Modeling Relational Data with Graph Convolutional Networks”, *The Semantic Web*, 2018.